

## Python Programming Assignment - 7

(1) what is the difference between local and global variables?

→ Local variable declared inside the function and it is accessible only inside that function.

Global variable declared outside function and Accessible everywhere but inside function we must understand scope.

(2) why should excessive use of global variable be avoided in large programs?

→ There many reasons to avoid

(1) Global variable degrade quality code

(2) Global variable create hidden connections.

(3) In large programme global variable travels many times and if some one uses same global variable name it will break the code.

(3) Explain the use of the global keywords. when should it be used?

→ variable inside a function belongs to the global scope outside the function, allowing you to modify.

(4) what is the special variable \_\_name\_\_ in python? who assigns it value?

→ \_\_name\_\_ is a built-in string variable in python that execution context of a script.

The Python interpreter automatically assigns the value to `--name--` at runtime, right before it begins executing any code in the file.

<5> what is the meaning of `--main--` in python execution?

→ when a python file is run directly, the value of `--name--` is set to "`--main--`".

`--main--` is special string that represent your programme start executing.

<6> Can function return another function? Explain conceptually?

→ yes function can return another function.

```
def hello_func(helloworld): # Parent function
```

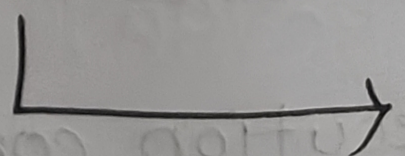
```
    def welcome_func(name) # child function
```

```
        return name
```

```
    return welcome_func # return child function.
```

<7> what is module in Python? How is it different from a function?

→ A module in python is simply a file containing python code variables, functions and classes that you can reuse in other python programs.



## module

<1> A module is the file & it contains function, variable

<2> A module contains function, classes, variable

<3> you access a module by importing.

<4> modules exist to organize large applications

## Function

<1> A function is a block-code written inside a function

<2> A function contains operational statements.

<3> Access a function by calling it.

<4> Functions exist to eliminate repetitive tasks.

<8> Explain what happens to \_\_name\_\_

- when a python file is executed directly.

- when the same file is imported as a module.

→ when a python file is run directly, the value of \_\_name\_\_ is set \_\_main\_\_ and entry point of execution is set main.

- when the same imported as a module the \_\_name\_\_ is set to module's file name.

<9> why do we write the condition if \_\_name\_\_ == "\_\_main\_\_":

→ we write this condition to control which code runs when a file is executed directly versus when it is imported into another file.

(10) Explain the difference between

- return x

- return x, y

- return

→ return x : This sends exactly one specific object back to the place where the function was called.

return x, y : (Return multiple values as a Tuple)

This allows a function to send back multiple values simultaneously.

return : This is a blank return statement & python automatically returns "None".  
(No value)